

# Anagha Prajapati

International Institute of Information Technology, Hyderabad

+91 9555357186 | [Email](#) | [LinkedIn](#) | [GitHub](#)

## EDUCATION

**International Institute of Information Technology, Hyderabad**

*B.Tech in Computer Science and Engineering*

**CGPA: 8.26/10**

*August 2024 – May 2028*

## WORK EXPERIENCE

**Undergraduate Researcher**

**January 2026 – Present**

*Software Engineering Research Center (SERC), IIIT-H*

*Hyderabad, India*

- Extending the EnCoDe framework by analyzing large-scale Python repositories to identify energy hotspots and quantify energy debt, building a context-aware routing framework that uses clustering to dispatch queries to optimal edge or cloud model based on energy efficiency and performance and researching self-adaptation for multi-agentic ML systems.

**Flutter Developer Intern**

**January 2026 – April 2026**

*Anjani Innovations*

*Hyderabad, India*

- Engineered an IoT-based remote patient vitals-monitoring system with BLE-enabled DHT11 and audio sensors on ESP32, a FastAPI + MongoDB backend, and a cross-platform Flutter app for live and historical monitoring.

## ACHIEVEMENTS

- Secured **AIR 1600** in **JEE Mains 2024** out of ~1.6 million candidates (**99.91%ile**)
- Secured **3<sup>rd</sup> Position** in Infinium's *Ctrl-Alpha: Stay Sharpe* quantitative trading challenge, IIIT-H
- **Runner-up**, MIT-style **Integration Bee**, IIIT-H

## PROJECTS

- **DNA Sequence Matching Analysis:** Implemented and benchmarked three DNA pattern-matching algorithms on genomic datasets, comparing search latency, memory usage, and robustness to identify optimal choices.
- **StatArb Pipeline:** Built a leakage-free, four-stage equity strategy pipeline on a 100-asset OHLCV dataset, causal feature engineering, walk-forward ensemble modeling, long/short backtesting, and a cointegration-based pairs-trading overlay.
- **Event Management System:** Built a full-stack MERN platform for IIIT-H's fest with role-based dashboards (Participant, Organizer, Admin), QR-based attendance tracking, real-time Socket.IO discussion forums, and a merchandise payment-approval workflow with transactional email notifications.
- **PEAD Research Chatbot:** Built a RAG chatbot for querying PEAD academic papers using Pinecone vector search, OpenAI embeddings, FAISS-based conversational memory with recency-weighted scoring, and GPT-4.1-mini, with citation-backed responses and dynamic PDF upload/delete management.
- **BookReads:** Developed a full-stack bookstore with browsing, cart and checkout, Firebase authentication, and a JWT-secured admin dashboard for inventory management and sales analytics with data visualizations.
- **Websitor:** Built an AI website generator powered by Google Gemini that converts natural-language prompts into responsive, animated single-file HTML sites, with a live preview and an in-browser Monaco code editor.
- **NexMeet:** Built a Zoom-style video conferencing platform supporting instant/scheduled meetings, personal rooms, recordings, and real-time in-call collaboration with screen sharing and a responsive Tailwind UI.
- **C-Shell:** Designed a custom Unix-like shell in C supporting built-in commands, I/O redirection, piping, signal forwarding, persistent command history, and foreground/background process and job control.
- **Network File System:** Built a distributed file system with a client-server architecture supporting remote file create/read/write/delete operations with consistency and fault tolerance.

## TECHNICAL AND SOFT SKILLS

- **Programming Skills:** C, C++, Python, JavaScript, TypeScript, SQL, HTML/CSS
- **Software Skills:** React.js, Next.js, Node.js, Express.js, FastAPI, Flutter, MongoDB, Tailwind CSS, Socket.IO, RESTful APIs, Git/GitHub, Linux/Unix
- **Machine Learning:** Pandas, NumPy, scikit-learn, XGBoost, Matplotlib, FAISS
- **Tools & APIs:** Pinecone, OpenAI API, Clerk, Stream Video SDK, Google Gemini API, Firebase, Arduino IDE, Postman, VS Code
- **Soft Skills:** Sports Representative, Sports Council, IIIT-H (August 2025 – Present); Finance Team Member, Arts Society (ArtSoc), IIIT-H

RELEVANT COURSEWORK

---

- |                                                                                                                                       |                                                                                                                           |                                                                                                                                |                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Data Structures</li><li>• Advanced Algorithm Design</li><li>• Data and Applications</li></ul> | <ul style="list-style-type: none"><li>• Design and Analysis of Software Systems</li><li>• Operating Systems and</li></ul> | <ul style="list-style-type: none"><li>Networks</li><li>• Probability and Statistics</li><li>• Machine, Data Learning</li></ul> | <ul style="list-style-type: none"><li>• Linear Algebra</li><li>• Data Visualization</li></ul> |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|